Recap

```
type(10)
```

```
int
```

```
type(4.6)
```

```
float
```

```
type("a")
```

```
str
```

```
type(True)
```

```
bool
```

```
type(false)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_498/27329152.py in <cell
line: 0>() ----> 1 type(false)

NameError: name 'false' is not defined
```

Next steps: Explain error

```
type(False)
```

```
bool
```

```
4 != 6
```

```
True
```

```
5 == 5
```

```
True
```

```
(4 != 6 ) or (5 == 5)
```

```
True
```

```
(4 != 6 ) or (5 != 5)
```

```
True
```

```
(4 == 6 ) or (5 != 5)
```

```
False
```

```
(4 != 6 ) and (5 == 5)
```

```
True
```

```
(4 == 6 ) and (5 == 5)
```

```
False
```

```
not (4 != 6)
```

```
False
```

Decision Making

```
Syntax:

if condition:
statements...
```

```
#Given the age of the person, check if they are eligibile to vote.
#if age is atleast 20, then they are eligible, else not
```

```
age = 19

if age >= 20:
  print("Eligible to vote")
```

```
age = 22

if age >= 20:
  print("Eligible to vote")
```

```
Eligible to vote
```

```
age = 22

if age >= 20:
print("Eligible to vote")
```

```
 File "/tmp/ipykernel_498/2914820533.py", line 4
    print("Eligible to vote")
    ^
IndentationError: expected an indented block after 'if' statement on line 3
```

Next steps: Explain error

```
#Q2. Take the password from the user. If it matches "hello@123" then print "Welcome".
```

```
password = input()

if password == "hello@123":
  print("Welcome")
```

```
hello
```

```
password = input()

if password == "hello@123":
  print("Welcome")
```

```
hello@123
Welcome
```

```
#if-else


age = 19

if age >= 20:
  print("Eligible to vote")
else:
  print("Not Eligible")
```

```
Not Eligible
```

```
#if else


age = 22

if age >= 20:
  print("Eligible to vote")
else:
  print("Not Eligible")
```

```
Eligible to vote
```

```
# Q2. Take the password from the user.
# If it matches "hello@123" then print "Welcome".
# Else, print "Invalid Password"
```

```
password = input()
if password == "hello@123":
  print("Welcome")
else:
  print("Invalid Password")
```

```
hello
Invalid Password
```

```
password = input()
if password == "hello@123":
  print("Welcome")
else:
  print("Invalid Password")
```

```
hello@123
Welcome
```

```
if 2> 3:
  print("A")
else:
  print("B")

print("C")
```

```
B
C
```

```
if 2 < 3:
  print("A")
else:
  print("B")

print("C")
```

```
A
C
```

```
#elif

marks =>
marks < 30 --> "Low"
marks between 30 and 70 --> "Avg"
marks > 70 --> "High"


# using if-else only
```

```
marks = 20

if marks < 30:
  print("Low")
else:
  if (marks >= 30) and (marks <= 70):
    print("Avg")
  else:
    print("High")
```

```
Low
```

```
marks = 40

if marks < 30:
  print("Low")
else:
  if (marks >= 30) and (marks <= 70):
    print("Avg")
  else:
    print("High")
```

```
Avg
```

```
marks = 85

if marks < 30:
  print("Low")
else:
  if (marks >= 30) and (marks <= 70):
    print("Avg")
  else:
    print("High")
```

```
High
```

```
# else if ==> elif

if 2 > 3:
  print("A")
else:
  print("B")
```

```
B
```

```
# else if ==> elif

if 2 < 3:
  print("A")
else:
  print("B")
```

```
A
```

```
if 2 < 3:
  print("A")
elif 5 > 4:
  print("B")
else:
  print("C")
```

```
A
```

```
if 2 > 3:
  print("A")
elif 5 > 4:
  print("B")
else:
  print("C")
```

```
B
```

```
if 2 > 3:
  print("A")
elif 5 < 4:
  print("B")
else:
  print("C")
```

```
C
```

```
#elif

marks =>
marks < 30 --> "Low"
```

```
marks between 30 and 70 --> "Avg"
marks > 70 --> "High"
```

```
marks = 85

if marks < 30:
  print("Low")
else:
  if (marks >= 30) and (marks <= 70):
    print("Avg")
  else:
    print("High")
```

```
marks = 20
if marks < 30:
  print("Low")
elif (marks >= 30) and (marks <= 70):
  print("Avg")
else:
  print("High")
```

```
Low
```

```
marks = 40
if marks < 30:
  print("Low")
elif (marks >= 30) and (marks <= 70):
  print("Avg")
else:
  print("High")
```

```
Avg
```

```
marks = 90
if marks < 30:
  print("Low")
elif (marks >= 30) and (marks <= 70):
  print("Avg")
else:
  print("High")
```

```
High
```

```
marks = 30
if marks < 30:
  print("Low")
elif (marks >= 30) and (marks <= 70):
  print("Avg")
else:
  print("High")
```

```
Avg
```

```
#Coolest code:

marks = 30
if marks < 30:
  print("Low")
elif (marks >= 30) and (marks <= 70):
  print("Avg")
else:
  print("High")
```

```
#Coolest code:

marks = 30
if marks < 30:
  print("Low")
elif (marks <= 70):
  print("Avg")
else:
  print("High")
```

```
Avg
```

Quiz

```
1.

age = 25

if age >= 18:
    print("Vote")

Ans: Vote
```

```
city = "Pune"

if city == "Pune"
    print("Welcome to Pune")
```

```
 File "/tmp/ipykernel_498/4127865888.py", line 3
    if city == "Pune"
                     ^
SyntaxError: expected ':'
```

Next steps: Explain error

```
city = "Pune"

if city == "Pune":
    print("Welcome to Pune")
```

```
Welcome to Pune
```

```
Start coding or generate with AI.
```